Assignment 1

10 pts

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CSE Department
Computer Vision CMAI 480

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1. [1 pt] Given the following matrix A, calculate its null space using MATLAB's Singular Value Decomposition (SVD) function.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Hint 1: Use the following command: $[U, S, V] = svd(A)$. Note that the null space can be computed from the $SVD(A) = U \sum V^T$ by locating the last row of V^T or last column of V corresponding to the smallest singular value of Σ.

Hint 2: Check the obtained null space vector(s) by applying the Null space definition $A\vec{x} = \vec{0}$.

2. [1 pt] Compute the null space of the matrix A in Question 1 using MATLAB's Symbolic Math Toolbox (or the built-in $null$ function). Compare/elaborate on the obtained result relative to Question 1.
Hint**:** You can define a symbolic matrix using $A = sym([1\ 2\ 3;\ 4\ 5\ 6;\ 7\ 8\ 9])$ and use null(A) to obtain the exact symbolic null space.

3. [1 pt] Compute the rank of the matrix *A* in (1) above using MATLAB.
Hint: use the following command: $rank(A)$

4. [1 pt] Compute the determinant of the matrix *A* in (1) above using MATLAB.
Hint: use the following command: $det(A)$

5. [2 pts] Calculate and analyze the eigenspaces of a small image using MATLAB: Use the following MATLAB code to randomly generate a 10 by 10 grayscale image, and then perform the suggested tasks:

% Create a 10x10 grayscale image with random integer pixel values between 0 and 255
$image = randi([0, 255], 10, 10);$

% Display the image
$imshow(uint8(image));$

$title('10x10\ Grayscale\ Image');$

Tasks:

a) Eigenvalues & Eigenvectors: Compute the eigenvalues and eigenvectors of the $image$ matrix. Use MATLAB's eig function. It is important to check if the

matrix is non-singular (i.e., check its rank) to perform this task. Otherwise, move to task (b).

b) SVD vs. EVD: Perform SVD on the **image matrix *A*** (i.e., $SVD(A) = U \sum V^T$). Extract the singular values and the right singular vectors from the V matrix. Also, compute $B = A^T A$ and perform Eigenvalue Decomposition (EVD) on B using $[P, D] = eig(B)$.
   - Compare the singular values of Σ with the square roots of the sorted (descending order) eigenvalues of B.
   - Compare the right singular vectors (V) with the eigenvectors (P). They should be approximately the same (up to a sign change in columns). Note: The left singular vectors are related to the eigenvectors of AA^T not $A^T A$.

6. [1 pt] Derive in just 3 steps (i.e., utilize the cross product operation) the intersection of two lines l_1 and l_2, with l_1 passing through the points (4,0) and (0,3), and l_2 passing through the points (-2,8) and (8,10).

7. [2 pts] Given a conic $x^2 + y^2 + 4x + 2y - 29 = 0$,
   (a) Where does the tangent to this conic at the perimeter point x = (1, 4) intersect with the y-axis?
   (b) Find the coordinates of the intersection of the tangents to this conic at points x_1 = (1,4) and x_2 = (3, -4).

8. [1 pt] The intersection of a plane through a double cone results in a degenerate conic. This conic is composed of two lines *l* and *m*, with *l* passing through the points (-1,-2) and (2,4), and *m* passing through the points (1,-3) and (4,10). What is the description of this degenerate conic as a 3x3 matrix C in homogeneous representation?
   Hint: use the homogeneous equation of the degenerate conic $C = l.m^T + m.l^T$

Deliverables: Use MATLAB ('.m' Script) to complete this assignment, and turn in

- Script file ('*.m'): Use section breaks using a double percentage sign (%%) to cleanly divide your script so that each section corresponds to and solves a specific question.
- A single PDF document containing your published MATLAB code along with the generated outputs and figures. Simply run the *publish* function directly inside your Command Window (*publish('your_script_name.m', 'pdf')*), or:
   - Click on the Publish tab at the top of the ribbon.
   - Click the small drop-down arrow under the Publish icon and select Edit Publishing Options.
   - Look for the Output settings pane and change the Output file format from html to pdf.
   - Ensure that Show code is set to true (this is the default).
   - Click the Publish button. MATLAB will execute the code and generate the PDF in a newly created html subfolder inside your current directory.

   Note: Make sure you do not end your crucial variable calculations with a semicolon (;). If you suppress the output in the script, it will not print text results into the PDF document.